

1. An 80-year-old man comes to the office for a routine follow-up examination. He has been hospitalized four times during the past year for management of congestive heart failure. He follows a 2 g/d sodium-restricted diet. His blood pressure is 150/100 mm Hg; other vital signs are within normal limits. The lungs are clear to auscultation. On cardiac examination, no murmurs or gallops are heard. Serum studies show a potassium concentration of 3.8 mEq/L, total cholesterol concentration of 260 mg/dL, and triglyceride concentration of 300 mg/dL. These findings have been consistent during the past year. Which of the following is most likely to decrease this patient's risk for future hospitalization?

- ☐ A) Eating a diet restricted to 500 mg of sodium daily
- ☐ B) Eating a diet rich in potassium
- ☐ C) Improving control of blood pressure
- ☐ D) Improving control of cholesterol concentration
- ☐ E) Improving control of triglyceride concentration



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2. A 47-year-old woman comes to the physician because of a 4-month history of intermittent substernal chest pain with exertion. She describes the pain as pressure-like and says it is relieved with rest. She does not have shortness of breath, nausea, or sweating during the episodes. She says the episodes do not seem to be occurring more frequently or with less exertion. She has no history of serious illness and takes no medications. She does not smoke. Her pulse is 90/min, and blood pressure is 150/90 mm Hg. Cardiac examination shows no abnormalities. An ECG shows no abnormalities. The patient is reluctant to undergo cardiac catheterization. Which of the following is the most appropriate next step in management?

- ☐ A) 24-Hour ambulatory ECG monitoring
- ☐ B) Measurement of serum creatine kinase activity and troponin I concentration
- ☐ C) Echocardiography
- ☐ D) Exercise stress test
- ☐ E) MUGA scan
- ☐ F) Aspirin and metoprolol therapy



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3. A 58-year-old man comes to the office because of a 5-day history of elevated home blood pressure readings. He has a 10-year history of hypertension. Since treatment with lisinopril was initiated 10 years ago, his previous home and office blood pressure measurements have been less than 140/90 mm Hg. Ten days ago, the patient's dentist initiated an acetaminophen, amoxicillin, and naproxen regimen for an abscessed tooth. The patient currently is asymptomatic. Blood pressure measured today in the office is 160/95 mm Hg; remainder of vital signs are within normal limits. Physical examination shows no abnormalities. Which of the following is the most likely explanation for this patient's findings?

- ☐ A) Antagonistic effects on cytochrome P450
- ☐ B) Decreased prostaglandin synthesis
- ☐ C) Diminished absorption of lisinopril
- ☐ D) Excessive adrenergic stimulation
- ☐ E) Inflammation of the renal interstitium



4. A previously healthy 32-year-old woman comes to the physician because of mild headaches and weakness for 3 months. Her pulse is 80/min, and blood pressure is 150/105 mm Hg. Examination shows no other abnormalities. Serum studies show:

Na ⁺	143 mEq/L
Cl ⁻	98 mEq/L
K ⁺	2.4 mEq/L
HCO ₃ ⁻	33 mEq/L
Urea nitrogen	18 mg/dL
Creatinine	0.8 mg/dL

Which of the following is the most likely diagnosis?

- ☐ A) Addison disease
- ☐ B) Adrenal adenoma
- ☐ C) Adrenogenital syndrome
- ☐ D) Cushing syndrome
- ☐ E) Hyperaldosteronism
- ☐ F) Pheochromocytoma



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5. A 15-year-old boy is brought to the office by his mother because of a 2-day history of moderate joint pain, fever, headaches, fatigue, and a blotch over his right shoulder blade. The blotch began as a small red papule and gradually spread. The patient says he thinks he was bitten by a tick 1 week ago. His temperature is 38.2°C (100.8°F); other vital signs are within normal limits. Examination shows a 15-cm erythematous area over the right scapula. Which of the following is the most appropriate next step in patient care?
- ☐ A) Serum titer of antibody to *Borrelia burgdorferi*
 - ☐ B) Serum titer of *Proteus vulgaris* Ox-19
 - ☐ C) Skin biopsy to identify microorganisms
 - ☐ D) Doxycycline therapy
 - ☐ E) Penicillin therapy



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6. A 65-year-old man is brought to the physician because of a 4-day history of increasingly severe upper back pain, pain with urination, nausea, fever, and chills. During the past 2 days, he has been able to ingest only clear fluids. He has a 5-year history of benign prostatic hyperplasia and a 25-year history of hypertension. His medications are tamsulosin and lisinopril. He is alert and cooperative. His temperature is 38.5°C (101.3°F), pulse is 96/min and regular, respirations are 18/min, and blood pressure is 104/68 mm Hg. Examination shows dry mucous membranes. Abdominal examination shows no abnormalities. There is costovertebral angle tenderness bilaterally. Which of the following is the most appropriate next step in management?

- ☐ A) Intramuscular administration of an antibiotic followed by a course of oral antibiotic therapy
- ☐ B) Intravenous administration of antibiotics and CT scan of the abdomen
- ☐ C) Intravenous administration of fluids and antibiotics
- ☐ D) Oral antibiotic therapy and re-examination in 2 days
- ☐ E) Oral antibiotic therapy and re-examination in 1 week



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7. A 36-year-old woman comes to the physician as a new patient for a follow-up examination because of ongoing “disabling” back pain. She rates her pain as a 10 on a 10-point scale. Physical examination and x-rays obtained during the past 3 years have shown no abnormalities. Back hygiene education, physical therapy, and acetaminophen and nonsteroidal anti-inflammatory drugs have not provided adequate relief. She believes that “something terrible is going on” despite being told by an orthopaedic surgeon that there is no significant pathology. She reports that she has seen a psychiatrist a few times but found the visits unhelpful. She has no other history of serious illness and currently takes no medications. She appears anxious and expresses heightened concern about her inability to stop her pain. Vital signs are within normal limits. Physical examination shows no abnormalities. Which of the following is the most appropriate next step in management?
- ☐ A) Application of a lidocaine patch
 - ☐ B) Oral gabapentin therapy
 - ☐ C) Oral oxycodone therapy
 - ☐ D) Oral sertraline therapy
 - ☐ E) Schedule frequent follow-up examinations



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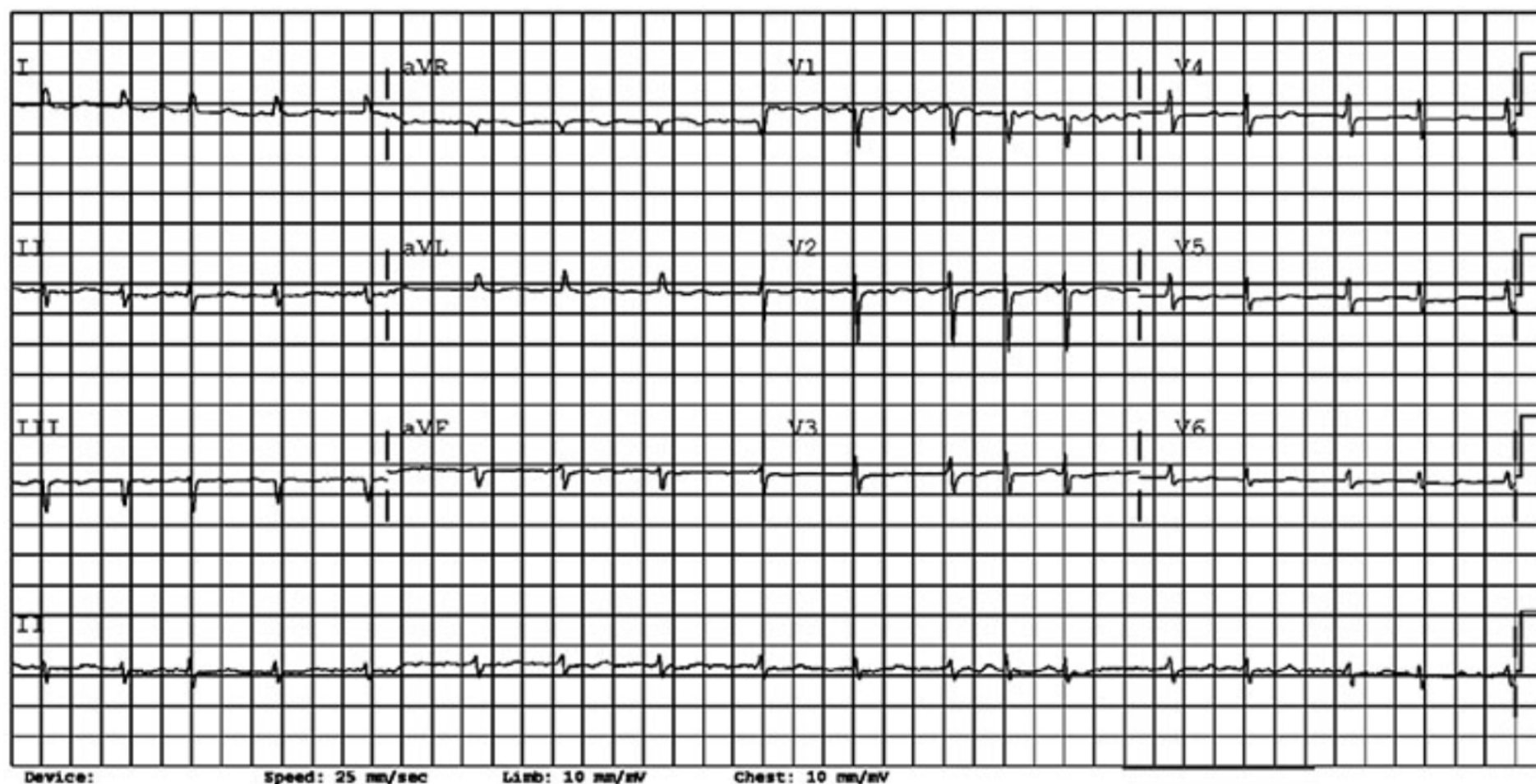
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8. A 78-year-old woman is brought to the office 2 hours after the sudden onset of shortness of breath, light-headedness, palpitations, and fatigue. She has a history of asymptomatic heart failure and hypertension. Her medications are lisinopril and aspirin. Her temperature is 37°C (98.6°F), pulse is 108/min, respirations are 24/min, and blood pressure is 100/60 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 92%. Bilateral basilar crackles are heard. An S₃ also is heard on cardiac examination. There is edema of the lower extremities. No other abnormalities are noted. Echocardiography 2 months ago showed an ejection fraction of 35%. ECG obtained 2 months ago showed no abnormalities. ECG obtained today is shown. Which of the following is the most likely explanation for these findings?

- ☐ A) Cerebral thromboembolism
- ☐ B) Decreased left ventricular filling
- ☐ C) Diastolic dysfunction
- ☐ D) Increased afterload
- ☐ E) Increased venous tone

9. A 77-year-old woman comes to the physician with her son for a follow-up examination. Two weeks ago, she was discharged from the hospital, where she was treated for chest pain and dyspnea and was diagnosed with left ventricular dysfunction. She is currently asymptomatic and is able to walk up three flights of stairs without symptoms. She has hypertension and hyperlipidemia. Before hospital admission, her medications were atenolol, simvastatin, aspirin, and calcium supplementation. On discharge, lisinopril and furosemide were added to the medication regimen. She lives alone. Her pulse is 68/min, and blood pressure is 132/90 mm Hg. Examination shows no abnormalities. The patient's son is concerned that his mother is confused by her medication regimen. Which of the following is the most appropriate next step in management?
- ☐ A) Ask the patient to bring in all current medications for review
 - ☐ B) Contact the hospital pharmacy to reconfirm all pre- and postadmission medications and dosages
 - ☐ C) Contact the patient's local pharmacy to confirm all preadmission medications and dosages
 - ☐ D) Recommend purchase of a daily medication storage and dispensing device
 - ☐ E) Review the medication list on the clinic chart



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10. A 62-year-old woman comes to the physician because of a lesion on her right cheek; she has had the lesion for 1 year but it has gradually increased in size over the past 6 months. She has a history of heavy sun exposure. Examination shows a 12-mm lesion on the right cheek with a rolled, pearly border, telangiectasia, and a small central crust. Which of the following is the most likely diagnosis?

- ☐ A) Basal cell epithelioma
- ☐ B) Keratoacanthoma
- ☐ C) Melanoma
- ☐ D) Metastatic cancer
- ☐ E) Squamous cell carcinoma



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11. A 40-year-old man has had fatigue for 2 weeks and progressively severe shortness of breath over the past 24 hours. He had been well until 3 weeks ago when he developed fever, sore throat, and rhinitis. Examination shows distended neck veins and basilar crackles. S_1 and S_2 are normal; an S_3 gallop is heard with no murmurs. Which of the following is the most likely diagnosis?

- ☐ A) Acute myocardial infarction
- ☐ B) Acute rheumatic fever
- ☐ C) Hypertrophic obstructive cardiomyopathy
- ☐ D) Pericarditis
- ☐ E) Viral myocarditis



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12. A 26-year-old man comes to the office for a health maintenance examination. He describes drinking at least six beers nightly during the past 2 years. He has no history of serious illness and takes no medications. Vital signs are within normal limits and physical examination discloses no abnormalities. He is concerned about his health and is thinking about quitting drinking alcohol, but he is not sure if he is ready. Serum laboratory studies show:

Alkaline phosphatase	65 U/L
AST	80 U/L
ALT	75 U/L

Which of the following strategies is most likely to motivate this patient to change his behavior?

- ☐ A) Acknowledge the discrepancy between his drinking and his desire to quit
- ☐ B) Convince the patient to accept the label that he is an alcoholic
- ☐ C) Describe the negative consequences of not following clinical advice
- ☐ D) Provide direction with a voice of authority
- ☐ E) Use humor whenever possible to lighten the mood



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13. A 29-year-old man comes to the physician because of a 3-week history of increasingly severe right shoulder pain. He says the pain begins as an ache at the front of his right shoulder and occasionally radiates down the front of his arm. He works as a lawyer and plays softball every weekend on his firm's team. He lifts weights at the gym. He sleeps well and is able to perform his usual work activities. Examination shows tenderness to palpation over the right bicipital groove. Flexion at the right elbow against resistance produces pain. Which of the following is the most appropriate initial step in management?
- ☐ A) 8-Day course of prednisone therapy
 - ☐ B) MRI of the right shoulder
 - ☐ C) Physical therapy
 - ☐ D) Recommendation for rest from softball and weight lifting
 - ☐ E) Sling immobilization of the right upper extremity for 3 days



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14. A 56-year-old man with hypertension comes to the office for a routine examination. He says he feels well. His only medication is lisinopril. He follows a low-cholesterol diet and walks 30 minutes daily. His temperature is 37.0°C (98.6°F), pulse is 75/min, respirations are 14/min, and blood pressure is 120/80 mm Hg. He is 183 cm (6 ft) tall and weighs 79 kg (175 lb); BMI is 24 kg/m². Examination shows no abnormalities. Laboratory studies are shown:

Cholesterol, total	194 mg/dL
HDL-cholesterol	35 mg/dL
LDL-cholesterol	135 mg/dL
Triglycerides	120 mg/dL

His estimated 10-year atherosclerotic cardiovascular disease risk is calculated at 8.3%. Addition of which of the following is the most appropriate next step in pharmacotherapy to prevent cardiovascular mortality in this patient?

- ☐ A) Atorvastatin
- ☐ B) Ezetimibe
- ☐ C) Gemfibrozil
- ☐ D) Niacin
- ☐ E) Omega-3 fish oil
- ☐ F) No change in pharmacotherapy is indicated



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15. A 77-year-old man is brought to the physician as a new patient by his wife and adult children. He received the diagnosis of dementia, Alzheimer type, 1 year ago and is aware of his diagnosis. His family reports that he enjoys gardening and reading but is unable to manage his finances. He has no other serious medical illnesses. He has not felt depressed. Vital signs are within normal limits. He is alert and is oriented to person and place but not to time. His Mini-Mental State Examination score is 24/30. The remainder of the examination shows no abnormalities. An advance directive has not been established. Which of the following is the most appropriate next step in discussing advance care planning?

- ☐ A) Advise the patient's wife to discuss the issues with the patient, then inform the physician of the decisions made
- ☐ B) Advise the patient's wife to obtain legal guardianship through a court order
- ☐ C) Ask the patient's children to establish the goals of treatment
- ☐ D) Discuss the issues in private with the patient's wife
- ☐ E) Discuss the issues with the patient



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16. A 64-year-old African American woman with type 2 diabetes mellitus comes to the office for a routine follow-up visit. The patient is concerned about increasingly dry skin on her feet. She takes metformin, atorvastatin, and enalapril. She has been trying to improve her diet and overall adherence to her diabetic treatment plan by following a carbohydrate counting plan. She is 163 cm (5 ft 4 in) tall and weighs 102 kg (225 lb); BMI is 39 kg/m². Vital signs are within normal limits. Physical examination of the feet shows linear, white, scaly areas with moist, raw skin between the toes. A photograph of the left foot is shown. Which of the following is the most appropriate treatment for this patient's rash?

- ☐ A) Clotrimazole cream
- ☐ B) Hydrocortisone ointment
- ☐ C) Mupirocin cream
- ☐ D) Petrolatum-based ointment
- ☐ E) Thorough washing of the feet twice daily



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17. A 19-year-old woman comes to the office because of a 4-hour history of severe right lower abdominal pain, heavy vaginal bleeding, light-headedness, and dizziness. Her last menstrual period was 2 months ago. She has no history of serious illness and takes no medications. She has been sexually active with one male partner for 2 years; they use condoms inconsistently. Her temperature is 37.0°C (98.6°F), pulse is 112/min, and respirations are 18/min. While supine, her blood pressure is 92/58 mm Hg; while sitting, her blood pressure is 76/52 mm Hg. Abdominal examination shows moderate tenderness to palpation of the right lower quadrant but no rebound. Pelvic examination shows a small pool of blood in the vaginal vault, a uterine size at the upper limit of normal, and moderate tenderness to palpation of the right adnexa. No other abnormalities are noted. Urine β -hCG testing is positive. Which of the following is the most likely diagnosis?

- ☐ A) Appendicitis
- ☐ B) Ectopic pregnancy
- ☐ C) Endometriosis
- ☐ D) Endometritis
- ☐ E) Threatened abortion



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18. A 20-year-old college student comes to the student health service because of insomnia, nausea, indigestion, and periodic anorexia over the past year. She has had a 4.5-kg (10-lb) weight loss over the past month. She worries about her physical problems and feels that her friends are avoiding her because she is not worthy of their friendship. She states that her grades have fallen, and she has dropped out of her sorority. She denies suicidal ideation but states that she cries daily for no apparent reason. She is 163 cm (5 ft 4 in) tall and weighs 45 kg (99 lb); BMI is 17 kg/m². Her pulse is 68/min, and respirations are 16/min. Examination and laboratory studies show normal findings. Which of the following is the most likely diagnosis?

- ☐ A) Adjustment disorder with depressed mood
- ☐ B) Anorexia nervosa
- ☐ C) Dysthymic disorder
- ☐ D) Major depressive disorder
- ☐ E) Somatization disorder



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19. A 32-year-old woman comes to the physician because of a 3-month history of diarrhea and rash on her knees. During this time, she also has had a 3.2-kg (7-lb) weight loss despite a normal appetite. She says that 1 to 2 hours after almost every meal, she has severe “rumbling and gurgling noises in my belly” and diarrhea containing partially undigested food. After the diarrhea begins, she notices that small, itchy, painless “red spots” appear on her knees. She has not had joint pain. She has not traveled recently. She is 165 cm (5 ft 5 in) tall and weighs 52 kg (115 lb); BMI is 19 kg/m². Her temperature is 36.2°C (97.1°F), pulse is 78/min, respirations are 16/min, and blood pressure is 112/62 mm Hg. Examination shows mildly pale conjunctivae. There are four areas of 5-mm, erythematous vesiculopapular lesions over the knees. The remainder of the examination shows no abnormalities. Laboratory studies show:

Hemoglobin	10.8 g/dL
Mean corpuscular volume	79 μm^3
Erythrocyte sedimentation rate	16 mm/h
Serum ferritin	10 ng/mL

Test of the stool for occult blood is negative. Which of the following is the most likely mechanism of these clinical findings?

- ☐ A) Absence of gastric intrinsic factor
- ☐ B) Granulomatous inflammation restricted to the intestinal mucosa
- ☐ C) Granulomatous inflammation through the intestinal mucosa
- ☐ D) Lactase deficiency
- ☐ E) Presence of anti-tissue transglutaminase antibody



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20. A 55-year-old woman comes to the physician for a health maintenance examination. She has smoked one pack of cigarettes daily for 35 years. Previous annual Pap smears have shown normal findings. It has been 5 years since her last mammography. She has never had a screening sigmoidoscopy or a test of the stool for occult blood. Her sister died of cervical cancer at the age of 50 years. This patient is at greatest risk for death from which of the following cancers?

- ☐ A) Breast
- ☐ B) Cervical
- ☐ C) Colon
- ☐ D) Lung
- ☐ E) Uterine



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21. A 4-year-old boy is brought to the physician 5 minutes after choking on a piece of meat. On arrival, he is coughing frequently but is able to vocalize. He is not cyanotic. His pulse is 145/min. Breath sounds are loud and sonorous on auscultation of the chest. Which of the following is the most appropriate next step in management?

- ☐ A) Back blows
- ☐ B) Chest compressions
- ☐ C) Heimlich maneuver
- ☐ D) Mouth-to-mouth resuscitation
- ☐ E) No immediate intervention necessary



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22. A 72-year-old woman comes to the physician because the physician's office was notified of one of the patient's lapsed prescription refills 1 month ago. She has a 22-year history of hypertension, 10-year history of osteoarthritis, and 8-year history of congestive heart failure. Her medications are lisinopril, carvedilol, furosemide, and acetaminophen; the dosage schedule of some of these medications requires that she take them several times daily. She lives alone and has no relatives near her home. She appears alert and in no distress. She is dressed appropriately and maintains eye contact. Vital signs are within normal limits. Examination shows mild osteoarthritic changes in the joints of the hands and knees. The remainder of the examination shows no abnormalities. Assessment of which of the following would be most effective in determining this patient's ability to manage her medications?

- ☐ A) Gait and balance
- ☐ B) Home environment
- ☐ C) Instrumental activities of daily living
- ☐ D) Likelihood of depression
- ☐ E) Social support



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23. A 77-year-old man is brought to the office by his daughter because of a 1-month history of progressive difficulty with memory. The patient moved to the USA from a small island in Micronesia 3 months ago to live with his daughter after his wife died. His daughter says that his memory was excellent 3 months ago but that it has declined noticeably during the past month. The patient has no history of serious illness and takes no medications. Vital signs are within normal limits. Physical examination shows no abnormalities. On mental status examination, the patient answers simple questions slowly but appropriately. He defers to his daughter to answer more complicated questions. He has a flat mood and affect. He recalls zero of three objects after 5 minutes. Which of the following is the most likely diagnosis?

- ☐ A) Creutzfeldt-Jakob disease
- ☐ B) Dementia, Alzheimer type
- ☐ C) Pseudodementia
- ☐ D) Vascular dementia
- ☐ E) Vitamin B₁ (thiamine) deficiency



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24. A 35-year-old woman with hypertension comes to the physician because of moderate right-sided chest pain and cough productive of blood that began suddenly 2 hours ago. During this time, she also has had mild pain and swelling of her right calf. Her medications are hydrochlorothiazide and an oral contraceptive. She has smoked one pack of cigarettes daily for 15 years. She is in mild respiratory distress. Her temperature is 37.4°C (99.3°F), pulse is 110/min and regular, respirations are 22/min, and blood pressure is 140/96 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 92%. Cardiopulmonary examination shows no abnormalities. Examination of the right calf shows warmth, edema, and tenderness to palpation. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Cardiac MRI
- ☐ B) CT angiography of the chest
- ☐ C) ECG
- ☐ D) PET scan
- ☐ E) Pulmonary function testing



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25. A 42-year-old woman comes to the physician because of a 10-month history of intermittent diffuse abdominal pain. The pain has occurred more frequently during the past month and is not associated with any other symptoms or changes in stool color. She has not had diarrhea or loose stools during the past month. She has not noticed any precipitating factors but notes that she has four children and is always under stress. She has a 10-year history of constipation and often does not have a bowel movement for 4 or 5 days. She has no history of serious illness and takes no medications. Last year, she was evaluated for similar symptoms; colonoscopy and CT scan of the abdomen and pelvis showed no abnormalities. Examination today shows tenderness over the left lower abdomen with palpable fullness in the lower left quadrant. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) 72-Hour collection of stool for measurement of fecal fat
- ☐ B) Psychiatric assessment
- ☐ C) Measurement of serum carcinoembryonic antigen concentration
- ☐ D) Serum antigliadin antibody assay
- ☐ E) Schilling test
- ☐ F) Biopsy of the small bowel
- ☐ G) No further evaluation is indicated at this time



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26. A 32-year-old man comes to the physician for a routine examination in late winter. He has a history of allergic rhinitis every spring and routinely takes diphenhydramine 2 to 3 times daily when his symptoms begin. His symptoms have been more severe during the past 2 years; he also has had sinusitis during this time requiring prolonged use of antibiotics. He is otherwise healthy and takes no medications. Examination shows no abnormalities. Which of the following is the most appropriate recommendation to prevent a recurrence of his allergic symptoms this spring?

- ☐ A) Recommend minimizing outdoor activities
- ☐ B) Recommend continued use of over-the-counter antihistamines
- ☐ C) Recommend desensitization therapy
- ☐ D) Prescribe oral montelukast
- ☐ E) Prescribe oral prednisone
- ☐ F) Prescribe inhaled nasal fluticasone



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27. A 57-year-old woman comes to the physician for a follow-up examination. She was diagnosed with primary hypothyroidism 8 weeks ago. She is currently asymptomatic. Her only medication is levothyroxine. Vital signs are within normal limits. Examination shows no abnormalities. Serum studies show a thyroid-stimulating hormone concentration of 9.5 $\mu\text{U/mL}$. Which of the following is the most appropriate next step in pharmacotherapy?

- ☐ A) Add liothyronine to the medication regimen
- ☐ B) Continue the current levothyroxine dosage
- ☐ C) Decrease the levothyroxine dosage
- ☐ D) Discontinue levothyroxine
- ☐ E) Increase the levothyroxine dosage



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28. A 68-year-old man comes to the physician because of a 1-month history of increasingly severe pain in his legs. He describes the pain as an ache that radiates to both calves. The pain becomes increasingly severe when he walks more than three blocks and resolves with rest. He has an 8-year history of low back pain after a fall from a ladder. His only medication is ibuprofen. He has smoked one-half pack of cigarettes daily for 45 years. He drinks two 12-oz beers daily. His temperature is 37°C (98.6°F), pulse is 86/min, respirations are 12/min, and blood pressure is 138/92 mm Hg. Examination shows mild edema of the lower extremities. There is a small effusion over both knees but no erythema or tenderness to palpation. The shins are cool to the touch. Crossed straight-leg raising test is negative, and deep tendon reflexes are 2+ bilaterally. Pedal pulses are 1+ and equal bilaterally. Capillary refill time is 3 seconds in both feet. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Arthrocentesis of both knees
- ☐ B) Determination of ankle brachial indices
- ☐ C) MRI of the lumbar spine
- ☐ D) Venous Doppler ultrasonography
- ☐ E) X-rays of both knees



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29. A 35-year-old woman who has been a patient for several years comes to the office because a pregnancy test done at home was positive. The pregnancy is confirmed by physical examination. During the past 5 months, she had two visits to the emergency department for traumatic injuries, each of which she attributed to her own clumsiness. She is married and has four children. Reluctantly, she reports concern about her personal safety and, with great embarrassment, confesses that the emergency department visits were precipitated by her husband hitting and pushing her. She says that she is afraid he will hurt her if she leaves him or if he finds out she is pregnant again. He lost his job nearly 1 year ago. She says that he is abusive only when he has been drinking. She tearfully asks to be referred to someone who can terminate her pregnancy. Which of the following is the most accurate statement concerning the patient's situation?
- ☐ A) Her husband's behavior is a logical consequence of his job loss
 - ☐ B) Her pregnancy increases her risk for abuse
 - ☐ C) Hiding all liquor in the house will reduce her risk for abuse
 - ☐ D) Marital therapy is usually helpful in reducing the risk for violence
 - ☐ E) Termination of the pregnancy will increase her risk for abuse



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30. An 18-month-old girl is brought to the physician by her mother for a routine examination. She was born at 32 weeks' gestation by cesarean delivery because of maternal preeclampsia. Apgar scores were 5 and 6 at 1 and 5 minutes, respectively. She was in the neonatal intensive care unit for 2 days because of fetal lung immaturity; she progressed well and was discharged at 1 week of age. She was exclusively breast-fed until the age of 6 months, when pureed foods were added to her diet. She was weaned off breast milk at the age of 14 months; at that time, soft foods and whole milk, which she tolerates well, were added to her diet. She receives no medications. Results of a complete blood count and measurement of blood lead concentration at the age of 1 year were within the reference ranges. She appears healthy. She is at the 25th percentile for length, 5th percentile for weight, and 25th percentile for head circumference. Vital signs are within normal limits. Examination shows no abnormalities. Screening for which of the following deficiencies is most appropriate at this time?

- ☐ A) Calcium
- ☐ B) Folic acid
- ☐ C) Iron
- ☐ D) Vitamin B₁₂ (cobalamin)
- ☐ E) Vitamin D



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31. A 17-year-old boy is brought to the physician shortly after injuring his right ankle during a high school football game. He says that his ankle “turned in” while he was trying to make a tackle. He was able to limp off the field and can bear weight on the ankle. Examination of the right ankle shows moderate swelling over the area distal to the lateral malleolus. The area is tender to palpation. There is no bony point tenderness. Joint testing shows no instability. Which of the following is the most appropriate next step in management?

- ☐ A) Apply ice and advise periodic leg elevation
- ☐ B) Apply topical capsaicin cream to the tender area
- ☐ C) Immobilize the ankle in a short-leg walking cast
- ☐ D) Inject a cortisone and lidocaine mixture into the tender area
- ☐ E) Order x-rays of the ankle and foot



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32. A 61-year-old woman comes to the office in July for a routine examination. She has hypertension and major depressive disorder. Her medications are hydrochlorothiazide and citalopram. She does not smoke cigarettes. Her blood pressure is 122/78 mm Hg; other vital signs are within normal limits. Examination shows no abnormalities. Which of the following is the most appropriate vaccine at this time?

- ☐ A) Herpes zoster
- ☐ B) Human papillomavirus
- ☐ C) Influenza
- ☐ D) Meningococcal
- ☐ E) Pneumococcal



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33. A previously healthy 42-year-old man comes to the physician because of right low back pain that began 2 days ago after he cleaned his garage. The pain radiates to the right buttock and posterior aspect of the right lower extremity to below the knee. He rates the pain as an 8 on a 10-point scale. It is worse when he is sitting. He has not had any other neurologic symptoms. He is visibly in pain as he tries to climb onto the examination table. His temperature is 37°C (98.6°F), pulse is 92/min, respirations are 12/min, and blood pressure is 142/88 mm Hg. There is straightening of the lumbar lordosis and tenderness to percussion over L4-5. Muscle strength is 5/5, and deep tendon reflexes and sensation are intact and symmetric. Straight-leg raise testing is positive on the right at 40 degrees with pain radiating down the right leg. Straight-leg raise testing is positive on the left at 45 degrees with pain radiating down the right leg. Which of the following is the most appropriate next step in management?

- ☐ A) Bed rest for 3 days
- ☐ B) Epidural corticosteroid injection
- ☐ C) MRI of the lumbar spine
- ☐ D) Normal activity as tolerated
- ☐ E) Transcutaneous electrical nerve stimulation
- ☐ F) Use of a lumbar support brace



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34. A large, multisite family medicine practice is planning a point-of-care ultrasonography screening program for abdominal aortic aneurysms in men age 65 to 75 years with a history of smoking. A pilot study of 100 patients in the practice group found point-of-care ultrasonography had the following results:

Sensitivity	90%
Specificity	90%
Positive predictive value	36
Negative predictive value	99
Positive likelihood ratio	9.00
Negative likelihood ratio	0.11

The practice's prevalence of abdominal aneurysm was estimated at 6% for this age group and gender. If the actual practice prevalence of aortic aneurysm is 5%, which of the following would decrease?

- ☐ A) Negative likelihood ratio
- ☐ B) Negative predictive value
- ☐ C) Positive likelihood ratio
- ☐ D) Positive predictive value
- ☐ E) Sensitivity
- ☐ F) Specificity

35. A previously healthy 15-year-old girl is brought to the physician because of a 2-day history of runny nose and nonproductive cough and a 1-day history of irritation and tearing of both eyes. The patient's temperature is 36.7°C (98°F); other vital signs are within normal limits. Examination shows a small amount of crusty discharge over the eyelids. The conjunctivae are injected. There is no photophobia. Visual acuity is 20/20 bilaterally. There is a small amount of clear nasal discharge. The lungs are clear to auscultation. Which of the following is the most appropriate next step in management?
- ☐ A) Culture of the ocular discharge
 - ☐ B) Oral doxycycline therapy
 - ☐ C) Symptomatic treatment only
 - ☐ D) Topical acyclovir therapy
 - ☐ E) Topical sulfacetamide therapy



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36. A 4-year-old boy is brought to the office by his parents because of a 2-day history of increasingly severe right lower abdominal pain and nausea. He has not had fever, vomiting, or diarrhea. He receives no medications. He recently emigrated with his family from Libya to the USA. He appears to be in mild distress. His temperature is 37.2°C (99.0°F), pulse is 90/min and regular, respirations are 14/min, and blood pressure is 118/74 mm Hg. Abdominal examination shows tenderness to palpation of the right lower quadrant with no rebound, guarding, or masses. The remainder of the examination shows no abnormalities. Laboratory studies are shown:

Hemoglobin	8.2 g/dL (N=11.5–14.5)
Mean corpuscular hemoglobin	20 pg/cell (N=25–31)
Mean corpuscular volume	65 μm^3 (N=76–90)
Leukocyte count	8600/mm ³ (with a normal differential)

Urinalysis shows no abnormalities. Which of the following is most likely to confirm the underlying diagnosis?

- ☐ A) Abdominal ultrasonography
- ☐ B) Abdominal x-ray
- ☐ C) Blood smear
- ☐ D) Diagnostic laparoscopy
- ☐ E) Examination of the stool for ova and parasites



37. A 45-year-old man with hypertension and sleep apnea comes to the office for a follow-up visit 4 weeks after an evaluation for polydipsia and polyuria disclosed a nonfasting serum glucose concentration of 220 mg/dL, an LDL-cholesterol concentration of 200 mg/dL, and hemoglobin A_{1c} of 8.5%. His only medication is lisinopril. The patient was advised to begin using a non-invasive positive pressure airway machine three months ago but does not use the machine nightly as recommended. During the initial evaluation, the patient was counseled on lifestyle modifications including weight loss, diet, and exercise. He has smoked one pack of cigarettes daily for the past 30 years and works as a stockbroker. He is 178 cm (5 ft 10 in) tall and weighs 120 kg (265 lb); BMI is 38 kg/m². His blood pressure is 150/92 mm Hg; the remainder of vital signs are within normal limits. Physical examination shows no abnormalities. In addition to initiating metformin therapy, which of the following is the most appropriate recommendation for this patient?

- ☐ A) Adjustment of medication to achieve target blood pressure
- ☐ B) Adjustment of medication to achieve target hemoglobin A_{1c}
- ☐ C) Examination of the feet in the office every 3 to 4 months
- ☐ D) Ophthalmologic examination every 3 years



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38. A popular 17-year-old high school student witnessed an argument between her present boyfriend and a previous boyfriend. She then ran away from home with her best friend and used excessive amounts of alcohol for 3 days. Throughout this period, she maintained her honors schoolwork and her after-school job. Which of the following is the most likely diagnosis?

- ☐ A) Adjustment disorder
- ☐ B) Alcohol dependence
- ☐ C) Conduct disorder
- ☐ D) Panic disorder
- ☐ E) Separation anxiety disorder



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39. A previously healthy 12-year-old boy is brought to the physician because of a 1-week history of sore throat and fatigue. His temperature is 38.1°C (100.5°F); other vital signs are within normal limits. Examination shows enlarged tonsils and pharyngeal erythema and exudates. The anterior cervical lymph nodes are tender to palpation. A complete blood count shows leukocytosis and atypical lymphocytes. A rapid streptococcal test is negative. Which of the following is the most appropriate pharmacotherapy?

- ☐ A) Intramuscular ceftriaxone
- ☐ B) Oral acyclovir
- ☐ C) Oral amoxicillin
- ☐ D) Oral aspirin as needed for fever
- ☐ E) Viscous lidocaine gargle for pain



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40. A 13-year-old girl is brought to the office by her mother for a routine well-child examination. She began menstruating 1 year ago. She eats healthy food including vegetables. She plays tennis and soccer, sleeps well, and has a supportive network of friends. She is getting straight A's in school. The patient has a bright affect. Vital signs are within normal limits and physical examination shows no abnormalities. The physician provides counseling about accident prevention, drugs, safe sex, and use of helmets when riding a bicycle or scooter. Counseling about which of the following additional topics should be provided at this visit?

- ☐ A) Career planning and college preparedness
- ☐ B) Choking hazards and poisoning prevention
- ☐ C) Depression and suicide prevention
- ☐ D) Vitamin D and folic acid supplementation
- ☐ E) No additional counseling is necessary



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41. A 75-year-old man comes to the physician with his wife because of a 1-day history of severe lower abdominal pain, nausea, vomiting, and an inability to urinate. Six months ago, the patient’s wife began to notice that he was taking longer than usual to urinate and leaving drops of urine on the toilet bowl rim and floor. During the past 5 days, he has made frequent trips to the bathroom and often has needed to return shortly after voiding. He has type 2 diabetes mellitus, hypertension, and stage 2 chronic kidney disease. His medications are glipizide, glargine insulin, and amlodipine. Examination shows suprapubic tenderness. Laboratory studies show:

Serum	
Na ⁺	140 mEq/L
K ⁺	4.1 mEq/L
Cl ⁻	101 mEq/L
HCO ₃ ⁻	25 mEq/L
Urea nitrogen	23 mg/dL
Creatinine	3.4 mg/dL
Urine	
pH	6.3
Specific gravity	1.015
Blood	1+
Glucose	negative
Bilirubin	negative
Protein	30 mg/dL
Ketones	negative
Nitrites	1+
Leukocyte esterase	1+
Urobilinogen	0.2 mg/dL (N<2)

Which of the following is the most likely underlying cause of this patient’s current symptoms?

- ☐ A) Acute tubular necrosis
- ☐ B) Benign prostatic hyperplasia
- ☐ C) Dehydration
- ☐ D) Nephrolithiasis
- ☐ E) Progression of hypertensive nephropathy

42. A 17-year-old girl comes to the office for an annual well-child examination. Her last visit to the physician was 14 months ago, during which she received one dose of the quadrivalent human papillomavirus (HPV) vaccine. She has no history of serious illness and takes no medications. During the past 2 years, she has been sexually active with one male partner; they use condoms consistently. Examination shows no abnormalities. Which of the following is the most appropriate next step in management regarding the HPV vaccine?
- ☐ A) Readministration of the first dose of the vaccine today with a recommendation that the patient receive a second dose in 2 months and a third dose in 6 months
 - ☐ B) Administration of a second dose of the vaccine today with a recommendation that the patient receive a third dose in 4 months
 - ☐ C) Administration of the last dose of the vaccine today
 - ☐ D) No further management is indicated



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43. A 20-month-old boy is brought to the office by his mother because she is concerned about his speech. She says he babbles various sounds but makes no words. He was born at term to a 26-year-old primigravid woman following an uncomplicated pregnancy and spontaneous vaginal delivery. Apgar scores were 8 and 9 at 1 and 5 minutes, respectively. He has no history of serious illness and receives no medications. The patient's grandmother assists with child care while the mother works part-time at a clothing store. The patient lives in a house constructed in 1938. He is at the 35th percentile for length and weight. Vital signs are within normal limits. Physical examination shows no abnormalities. He is active and responds to aural and tactile stimuli. He has limited direct eye contact, makes babbling sounds, and does not say any words. Which of the following is the most likely diagnosis?

- ☐ A) Autism spectrum disorder
- ☐ B) Down syndrome
- ☐ C) Failure to thrive
- ☐ D) Hearing deficit
- ☐ E) Lead poisoning



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44. A 37-year-old man comes to the office because of a 1-week history of visual disturbances, flashing lights, and blurred vision. He tested positive for HIV infection 8 years ago. He has not been taking his antiretroviral medications during the past 4 months, explaining, "I just got tired of the daily routine." His temperature is 37.1°C (98.8°F), pulse is 84/min, respirations are 16/min, and blood pressure is 128/76 mm Hg. Physical examination discloses no abnormalities. Which of the following is the most likely infectious agent?

- ☐ A) *Cryptococcus neoformans*
- ☐ B) *Cryptosporidium parvum*
- ☐ C) Cytomegalovirus
- ☐ D) Rubella virus
- ☐ E) *Toxoplasma gondii*



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45. A previously healthy 7-year-old boy is brought to the physician because he started limping on his left leg 3 days ago and now refuses to bear weight on it. The mother says that he has “felt feverish” to touch. He has no history of recent trauma. He is at the 45th percentile for height and the 60th percentile for weight. His temperature is 38.9°C (102°F), pulse is 108/min, respirations are 14/min, and blood pressure is 110/72 mm Hg. The patient holds the left hip flexed, abducted, and externally rotated. Passive and active range of motion of the left hip is limited by pain. The left sacroiliac joint is nontender to palpation. Laboratory studies show a leukocyte count of 15,500/mm³, erythrocyte sedimentation rate of 110 mm/h, and a serum C-reactive protein concentration of 4.2 mg/L (N=0.08–3.1). Which of the following is the most likely diagnosis?

- ☐ A) Acute rheumatic fever
- ☐ B) Legg-Calvé-Perthes disease
- ☐ C) Septic arthritis
- ☐ D) Slipped capital femoral epiphysis
- ☐ E) Toxic transient synovitis



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46. A 67-year-old woman comes to the physician because of a 2-month history of fatigue and increased urine output. She has hypertension treated with enalapril. Her blood pressure is 143/88 mm Hg. Examination shows no other abnormalities. Laboratory studies show:

Serum	
Ca ²⁺	11.9 mg/dL
Phosphorus	2.1 mg/dL
Albumin	4.8 g/dL
Urine specific gravity	1.005

Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Glucose tolerance test
- ☐ B) Urine culture
- ☐ C) Measurement of serum parathyroid hormone concentration
- ☐ D) Measurement of serum vitamin D concentration
- ☐ E) Intravenous pyelography



47. A 22-year-old woman comes to the physician in January because of a 6-week history of an itching, burning rash over her forearms and wrists. She has a history of similar rashes around this time of year but says that this rash is the worst she has ever experienced. Two weeks ago, she had an upper respiratory tract infection that resolved spontaneously. She has no other history of serious illness and takes no medications. A photograph of the right upper extremity is shown. These findings also appear over the volar aspect of the wrists. The remainder of the examination shows no abnormalities. Which of the following is the most appropriate next step in management?

- ☐ A) Oral cyclosporine therapy
- ☐ B) Oral methotrexate therapy



47. A 22-year-old woman comes to the physician in January because of a 6-week history of an itching, burning rash over her forearms and wrists. She has a history of similar rashes around this time of year but says that this rash is the worst she has ever experienced. Two weeks ago, she had an upper respiratory tract infection that resolved spontaneously. She has no other history of serious illness and takes no medications. A photograph of the right upper extremity is shown. These findings also appear over the volar aspect of the wrists. The remainder of the examination shows no abnormalities. Which of the following is the most appropriate next step in management?

- ☐ A) Oral cyclosporine therapy
- ☐ B) Oral methotrexate therapy
- ☐ C) Topical ketoconazole cream
- ☐ D) Topical moisturizing cream
- ☐ E) Ultraviolet light therapy



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48. A 25-year-old man comes to the physician for evaluation 2 days after his sexual partner was diagnosed with acute hepatitis B. Examination shows no abnormalities. Three days later, laboratory studies show:

Hepatitis B surface antigen	negative
Antibody to hepatitis B surface antigen	negative
Antibody to hepatitis B core antigen	negative

In addition to condom use counseling, which of the following is the most appropriate recommendation to prevent hepatitis B in this patient?

- ☐ A) Hepatitis B immune globulin only
- ☐ B) Hepatitis B vaccine series only
- ☐ C) Hepatitis B immune globulin and hepatitis B vaccine series
- ☐ D) Corticosteroid therapy only
- ☐ E) Ganciclovir therapy only
- ☐ F) Azathioprine therapy only
- ☐ G) Azathioprine and corticosteroid therapy
- ☐ H) No preventative measures indicated



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49. A 47-year-old woman comes to the physician because of nonproductive, "bothersome" cough almost every day for 2 months, especially at night. Her symptoms have not responded to a 4-week course of antihistamines and decongestants. She has not had fever, shortness of breath, or any other symptoms. She has hypothyroidism well controlled with levothyroxine. There is no family history of serious illness. She has smoked one-half pack of cigarettes daily for 25 years. Her temperature is 37°C (98.6°F), pulse is 60/min and regular, respirations are 12/min, and blood pressure is 110/76 mm Hg. The oropharynx appears normal with no cobblestoning. Cardiopulmonary examination shows no abnormalities. Which of the following is the most likely cause of this patient's cough?

- ☐ A) Allergic rhinitis
- ☐ B) Chronic sinusitis
- ☐ C) Emphysema
- ☐ D) Gastroesophageal reflux disease
- ☐ E) Upper respiratory tract infection



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50. A 66-year-old man comes to the physician because of a 1-year history of shortness of breath with exertion and progressive cough productive of small amounts of white sputum. He has no history of serious illness and takes no medications. He has smoked one pack of cigarettes daily for 50 years. Vital signs are within normal limits. Pulse oximetry on room air shows a resting oxygen saturation of 91%; his oxygen saturation is 89% with ambulation. Pulmonary examination shows a prolonged expiratory phase. Prior spirometry shows a baseline FEV₁ of 55% of predicted and an FEV₁:FVC ratio of 60%. The FEV₁ improved by 15% after administration of a bronchodilator. Which of the following is most likely to prevent worsening of this patient's underlying condition?

- ☐ A) Combined inhaled corticosteroid and long-acting β -adrenergic agonist therapy
- ☐ B) Long-acting anticholinergic therapy
- ☐ C) Long-term, low-dose oral corticosteroid therapy
- ☐ D) Oxygen therapy
- ☐ E) Smoking cessation



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